

Mode 9 — Baseline-BFP

DP8 (16×16) → DP8 (16×8) → DP8 (16×4) → DP8 (8×4) → block exponents

Step 1 — DP8 (16×16)

$$\text{DP8 (16} \times \text{16)} = \sum_{i=0}^7 a_i b_i$$

Step 2 — DP8 (16×8)

$$\text{DP8 (16} \times \text{16)} = \underbrace{2^8 \sum_{i=0}^7 a_i b_i^{hi}}_{\text{DP8 (16} \times \text{8)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{lo}}_{\text{DP8 (16} \times \text{8)}}$$

Step 3 — DP8 (16×4)

$$\text{DP8 (16} \times \text{16)} = \underbrace{2^{12} \sum_{i=0}^7 a_i b_i^{hh}}_{\text{DP8 (16} \times \text{4)}} + \underbrace{2^8 \sum_{i=0}^7 a_i b_i^{hl}}_{\text{DP8 (16} \times \text{4)}} + \underbrace{2^4 \sum_{i=0}^7 a_i b_i^{lh}}_{\text{DP8 (16} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i b_i^{ll}}_{\text{DP8 (16} \times \text{4)}}$$

Step 4 — DP8 (8×4)

$$\begin{aligned} \text{DP8 (16} \times \text{16)} = & \underbrace{2^{20} \sum_{i=0}^7 a_i^{hi} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{2^{16} \sum_{i=0}^7 a_i^{hi} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{2^{12} \sum_{i=0}^7 a_i^{lo} b_i^{hh}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} \\ & + \underbrace{2^8 \sum_{i=0}^7 a_i^{lo} b_i^{hl}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{2^8 \sum_{i=0}^7 a_i^{hi} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{2^4 \sum_{i=0}^7 a_i^{lo} b_i^{lh}}_{\text{DP8 (8} \times \text{4)}} + \underbrace{\sum_{i=0}^7 a_i^{lo} b_i^{ll}}_{\text{DP8 (8} \times \text{4)}} \end{aligned}$$

Step 5 — Block exponents (one exponent block per DP8 (8×4) — all copies of one E ; parallel matmul: own E_a, E_b)

$$(P_j, E) \text{ per DP8 (8} \times \text{4) above,} \quad E = E_a + E_b \longrightarrow (M, E)$$

$\delta \equiv 0$: M is the Step-4 sum unchanged; $\text{DP8 (16} \times \text{16)} = M \cdot 2^E$.

Step 6 — Result

$$\begin{aligned} M = & 2^{20} \sum_{i=0}^7 a_i^{hi} b_i^{hh} + 2^{16} \sum_{i=0}^7 a_i^{hi} b_i^{hl} + 2^{12} \sum_{i=0}^7 a_i^{lo} b_i^{hh} + 2^{12} \sum_{i=0}^7 a_i^{hi} b_i^{lh} \\ & + 2^8 \sum_{i=0}^7 a_i^{lo} b_i^{hl} + 2^8 \sum_{i=0}^7 a_i^{hi} b_i^{ll} + 2^4 \sum_{i=0}^7 a_i^{lo} b_i^{lh} + \sum_{i=0}^7 a_i^{lo} b_i^{ll}, \quad E = E_a + E_b \end{aligned}$$